An Amplitude–Phase Informed Diffusion Model for Near-Range SAR Imaging With Single Frequency and Sparse Arrays

Lei Wang, Xianxun Yao, *Member, IEEE*, Tiancheng Song, Lening Tian, Ruijiao Tian, and Guolin Sun

***Abstract*—Reconstructing millimeter-wave (MMW) images from single-frequency, spatially sparse sampling measurements reduces system complexity and facilitates real-time sensing. However, due to the limited information in sparse measurements, existing reconstruction methods often struggle to recover high-quality images with precise edges, fine structural details, and effective artifact suppression. An amplitude–phase informed diffusion (APIDiff) model for sparse reconstruction is proposed in this article, where the amplitude information is exploited to characterize the object's 2-D planar shape and scattering intensity, while the phase information is leveraged to delineate object contours, textures, and 3-D structural characteristics. A spatial-wavelet (SW) encoder is introduced to extract amplitude and phase features from both spatial and frequency domains. A subsequent multimodal fusion module effectively integrates these features to enhance feature representation, providing conditional inputs to APIDiff. Additionally, an adaptive weighting strategy is integrated to accelerate model convergence. Extensive experiments on both simulated and real-world datasets demonstrate that APIDiff consistently outperforms existing methods in imaging quality across various spatial sampling rates (SRs) and signal-to-noise ratios (SNRs).**

***Index Terms*—Diffusion model, millimeter-wave (MMW) imaging, near-range imaging, phase feature, single-frequency imaging, sparse imaging, wavelet feature.**

I. Introduction

NEAR-FIELD millimeter-wave (MMW) imaging has been widely studied and applied across various fields, such as concealed object detection [1], [2], nondestructive testing [3], [4], through-wall imaging [5], [6], and biomedical imaging [7], [8], due to its strong penetration capability through most optically opaque materials and its nonionizing nature, which poses fewer health risks. By integrating synthetic aperture radar (SAR) techniques, MMW imaging can achieve high azimuth resolution through a large virtual aperture and high range resolution with a wideband signal [9], [10]. However, the above requirements significantly increase the system's complexity and cost, hindering its practical application. Reconstructing images from incomplete measurements, such as single frequency or sparse arrays, has emerged as a common strategy to reduce both measurement time and hardware costs.

Existing MMW imaging algorithms can be classified into three categories. The first category is based on matched filtering (MF), including time-domain methods such as the backprojection algorithm (BPA) [11] and frequency-domain methods such as the range migration algorithm (RMA) [12] and ωK algorithm [13]. These conventional methods provide a direct and effective solution and can generate high-quality SAR images under wideband and spatially dense sampling conditions. However, in the case of single-frequency and spatially sparse sampling, the lack of signal bandwidth eliminates range resolution, leading to image defocusing and blurring [14], while insufficient spatial sampling violates the Nyquist criterion, resulting in significant grating lobe artifacts [15], [16].

Compressed sensing (CS) theory [17] has been proven capable of reconstructing signals from highly incomplete measurements and is widely applied in MMW sparse reconstruction [18], where the signal acquisition and imaging process are modeled as a linear shift-invariant system and a rank-deficient inverse problem, due to the inherent sparsity and undersampling of the signal. The reconstruction problem is typically addressed within an optimization framework using standard algorithms, such as orthogonal matching pursuit (OMP) [19], alternating direction method of multipliers (ADMM) [20], iterative shrinkage-thresholding algorithm (ISTA) [21], and its accelerated variant, fast ISTA (FISTA) [22]. Considerable efforts have been dedicated to exploring CS-based MMW sparse reconstruction methods. For instance, the 3-D ωK algorithm is integrated with the CS principle in [23] for enhanced image reconstruction. Li et al. [24] proposed an interpolation-free sensing operator to circumvent computationally intensive matrix operations. Furthermore, Markarian and Ghofrani [25] replace the ℓ_1 regularization with high-order total variation (TV) to achieve improved speckle suppression and edge preservation. However, CS-based reconstruction methods encounter two fundamental limitations in imaging performance. The primary limitation stems from the reliance on manually designed sparsity priors, which exhibit limited representational capacity and are incapable of

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capturing the high-level structural and semantic characteristics of real targets, such as shapes, sharp edges, and delicate textures. The secondary limitation is their strong dependence on manually adjusted hyperparameters, which compromises robustness across diverse imaging scenarios [26].

The third category is deep learning (DL)-based methods, which enable the automatic extraction of hierarchical image features through training, generating richer and more semantically meaningful prior information, thereby significantly enhancing the quality of sparse reconstruction [27]. A direct approach utilizes deep neural networks (DNNs) to learn the nonlinear mapping from raw received signals to SAR images, demonstrated in [28] through a cascade of convolutional and fully connected layers. However, a large number of network parameters are required to model the high-dimensional nonlinear mapping from the signal domain to the image domain, and there are no theoretical guarantees. An alternative approach integrates DL networks with traditional methods, typically the deep unfolding networks (DUNs), such as LFIST-Net [29], SFH-ADMM-Net [30], CSR-Net [31], and LISTA [32], which unfold the CS-based methods into layer-fixed hierarchical flow graphs and learns the hyperparameters through an end-to-end training process, achieving significantly improved computational efficiency while maintaining comparable accuracy to the traditional CS-based methods. However, the prior information in SAR images is not further exploited in DUNs and remains limited to target sparsity, which constrains the modeling capability and hinders further improvements in reconstruction quality. Other studies [33] adopt a two-stage reconstruction framework. In the first stage, low-resolution or fuzzy images are reconstructed from the received signals with traditional MF-based methods. In the second stage, a pretrained DL network is utilized to improve the imaging quality by suppressing artifacts and recovering fine structural details. However, the final imaging performance is inherently constrained by the fidelity of the first-stage reconstruction, and these methods are susceptible to over-smoothing, compromising the accurate recovery of delicate image structures [34] when low-fidelity first-stage results are obtained.

Inspired by the remarkable performance of diffusion models in generative tasks, such as image synthesis [35], speech generation [36], and medical image reconstruction [37], we proposed a two-stage reconstruction method, the single-frequency autofocusing diffusion (SFADiff) model, in our previous work [14], which is the first to introduce conditional diffusion models into MMW near-field single-frequency imaging. In the second stage of SFADiff, a 3-D convolutional network is employed to extract amplitude features of defocused 3-D images obtained in the first stage, guiding the reverse process toward focused SAR images. However, SFADiff performs poorly in the case of single-frequency and spatially sparse sampling, due to its inherent reliance on the quality of the first-stage reconstructed images, where the amplitude features are severely degraded, resulting from the additional spatial sparsity constraints.

Based on our previous work, this article proposes an amplitude–phase informed diffusion (APIDiff) model for near-range SAR imaging with single-frequency and sparse array configurations. Unlike traditional methods that rely solely on degraded amplitude features to characterize the object's 2-D planar shape and scattering intensity, APIDiff fully exploits phase information as complementary features, enhancing the discriminative representation of objects. Specifically, phase continuity delineates object contours, enhancing contour modeling capability and boundary localization accuracy. The phase gradient encodes surface depth variations, providing critical cues for reconstructing object textures and 3-D structures. To effectively extract both amplitude and phase features, a spatial-wavelet (SW) encoder is proposed, which jointly leverages spatial- and frequency-domain information. Within the SW encoder, a residual-in-residual dense block (RRDB) is proposed to extract the spatial features, and the discrete wavelet transform (DWT) is employed to decompose the input image into four frequency subbands, which are then processed individually by separate RRDBs for frequency-domain feature extraction. The extracted features are fused through a multimodal fusion module and employed as conditions for APIDiff to provide informative object priors, enhancing the preservation of image details. Besides, an adaptive weighting strategy is introduced to accelerate model convergence by incorporating a trainable condition weight parameter.

In conclusion, the contributions of this article are listed as follows.

1) This article proposes a conditional diffusion model for reconstructing high-quality MMW images with single-frequency and spatially sparse sampling measurements.
2) This article conducts an in-depth analysis of phase features in SAR images and integrates both amplitude and phase features as conditional priors to guide the reverse process of the diffusion model.
3) To enhance feature representation, an SW encoder is proposed based on DWT, which can simultaneously extract spatial- and frequency-domain features. Besides, an adaptive weighting strategy is adopted to accelerate model convergence.
4) Extensive experiments on both simulated and real-world datasets demonstrate that the proposed method outperforms existing algorithms in imaging quality, while exhibiting strong robustness to noise and interference.

The remainder of this article is organized as follows. Section II presents limitations of traditional MMW imaging and introduces fundamental principles of diffusion models. Section III provides an in-depth analysis of phase features in SAR images and details the architecture of the proposed model, APIDiff, along with its training and inference procedures. Extensive simulations and real-world experiments are conducted in Section IV to validate the effectiveness of the proposed method. Finally, the conclusions are summarized in Section V.

II. Basic Theory

This section first reviews the limitations of traditional MMW imaging, including image defocusing and grating lobe artifacts, under the conditions of single-frequency and spatially

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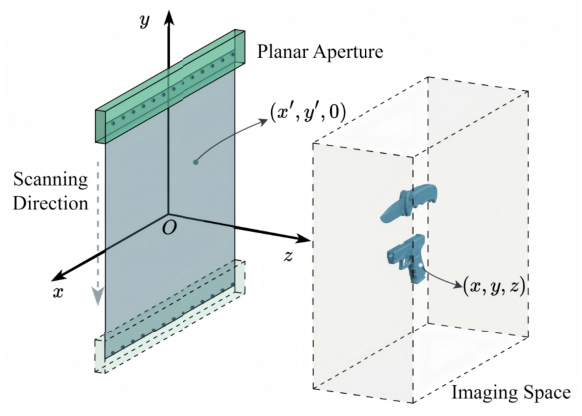

Fig. 1. Near-range SAR 3-D imaging geometry.

sparse sampling. Subsequently, the theoretical principles of diffusion models are introduced, establishing the proposed method's theoretical foundations.

A. Limitations of Traditional MMW Imaging

The geometry for the near-range SAR 3-D imaging system is illustrated in Fig. 1. A horizontal linear single-input-single-output (SISO) array is mechanically scanned in the vertical direction, forming a large rectangular synthetic aperture. During the scanning process, MMW signals are transmitted by the array's transmitting elements, scattered by targets within the imaging region, and subsequently received by the array's receiving elements, enabling effective target sensing. The coordinates of the array elements and the imaging region are denoted as $(x', y', 0)$ and (x, y, z), respectively.

Under the Born approximation and by neglecting the propagation attenuation, which is considered negligible in SAR imaging, the measured signal $s(x', y', k)$ can be expressed as

$$s\left(x', y', k\right) = \iiint f\left(x, y, z\right) \times e^{-j2k\sqrt{(x-x')^2+(y-y')^2+z^2}} dxdydz \tag{1}$$

where $k = 2\pi f/c$ denotes the wavenumber of the MMW signal, f is the signal frequency, and c is the speed of light. $f(\cdot)$ represents the reflectivity map of the target, which constitutes the ground truth of SAR images.

MF-based methods are direct and effective reconstruction solutions for measurements with sufficient bandwidth and spatial sampling. Among them, BPA, which is widely regarded as the gold standard, is formulated as

$$\hat{f}\left(x, y, z\right) = \iiint s\left(x', y', k\right) \times e^{j2k\sqrt{(x-x')^2+(y-y')^2+z^2}} dx'dy'dk. \tag{2}$$

Moreover, its frequency-domain variant, RMA, is expressed as

$$\hat{f}\left(x, y, z\right) = \mathcal{F}_{3-D}^{-1}\left[\text{Interp}\left[\mathcal{F}_{2-D}\left[s\left(x', y', k\right)\right]\right]\right] \tag{3}$$

where $\mathcal{F}_{2-D}[\cdot]$ and $\mathcal{F}_{3-D}^{-1}[\cdot]$ denote the 2-D Fourier transform and 3-D inverse Fourier transform, respectively. Interp[·] represents the Stolt interpolation [9].

By performing the Fourier transform and inverse Fourier transform, (3) can be simplified as

$$\begin{aligned} \hat{F}\left(k_x, k_y, k_z\right) &= S\left(k_x, k_y, k\right) \\ k_z &= \sqrt{4k^2 - k_x^2 - k_y^2} \end{aligned} \tag{4}$$

where k_x, k_y, and k_z are the frequency-domain counterparts of x, y, and z, respectively. $\hat{F}(\cdot)$ and $S(\cdot)$ denote the spectra of $\hat{f}(\cdot)$ and $s(\cdot)$, respectively.

The above equation establishes a fundamental relationship between the signal spectrum and the imaging result, enabling imaging quality assessment through analysis of the signal spectrum support, as detailed in our previous work [10], [16]. Under conditions that satisfy the Nyquist sampling, the spectrum of the imaging result for a single point scatterer at (x_i, y_j, z_l) is summarized as

$$\begin{aligned} \hat{F}\left(k_x, k_y, k_z\right)\big|_{(x_i,y_j,z_l)} &= e^{-jk_x x_i - jk_y y_j - jk_z z_l} \\ &\quad \times \frac{k^2 z_l \left(z_{\max}^2 - z_{\min}^2\right)}{k_z^4} \\ &\quad \times \text{Supp}\left(k_x, k_y, k_z\right)\big|_{(x_i,y_j,z_l)} \end{aligned} \tag{5}$$

where Supp(·) denotes the spectrum support, formulated as

$$\begin{aligned} &\text{Supp}\left(k_x, k_y, k_z\right)\big|_{(x_i,y_j,z_l)} \\ &= \begin{cases} 1, & \text{if} \begin{cases} 2k_{\min} \le \sqrt{k_x^2 + k_y^2 + k_z^2} \le 2k_{\max} \\ \arctan\dfrac{x_i - x'_{\max}}{z_l} \le \arctan\dfrac{k_x}{k_z} \le \arctan\dfrac{x_i - x'_{\min}}{z_l} \\ \arctan\dfrac{y_j - y'_{\max}}{z_l} \le \arctan\dfrac{k_y}{k_z} \le \arctan\dfrac{y_j - y'_{\min}}{z_l} \end{cases} \\ 0, & \text{otherwise} \end{cases} \end{aligned} \tag{6}$$

where $k_{\min}$ and $k_{\max}$ are the bounds of the signal wavenumber. $x'_{\min}$, $x'_{\max}$ and $y'_{\min}$, $y'_{\max}$ represent the horizontal and vertical extents of the synthetic aperture, respectively.

According to (5) and (6), under Nyquist sampling conditions, the imaging result's spectrum is confined to a spherical sector region, with radius determined by the signal bandwidth and angular extent defined by the spatial azimuth angles.

Fig. 2(a) illustrates the spectrum under wideband and spatially dense sampling conditions. The wideband signal and large aperture ensure sufficient spectral extent along all three dimensions, while the dense sampling prevents spectral aliasing, leading to a high-quality imaging result, as shown in Fig. 2(e). To reduce the system complexity, the single-frequency measurement is commonly adopted, which degrades the spectrum support to a spherical shell, as depicted in Fig. 2(b). The absence of spectral extent along the depth dimension results in degraded range resolution and ultimately causes defocusing, as demonstrated in Fig. 2(f). Furthermore, the spectra and imaging results under single-frequency and spatially sparse sampling conditions are shown in Fig. 2(c) and (g) and (d) and (h) for uniform and random sampling, respectively. These results exhibit prominent grating lobe artifacts resulting from the spectrum aliasing due to the violations of the Nyquist criterion. Moreover, the energy of the

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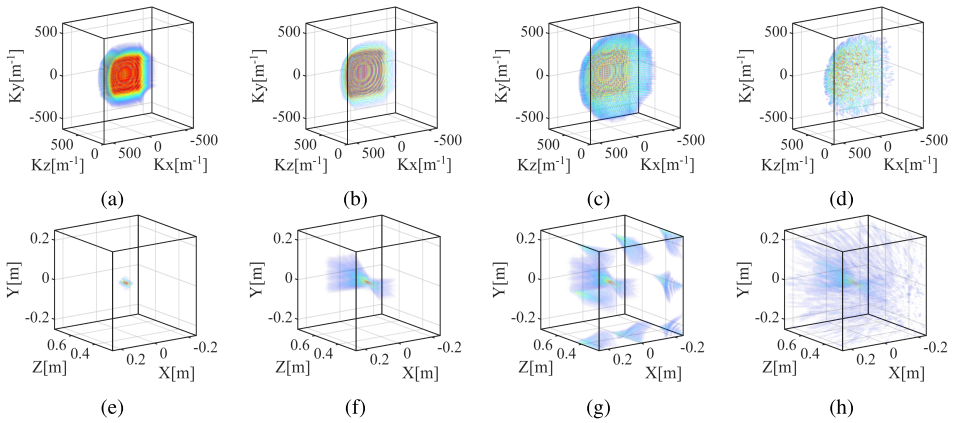

Fig. 2. Comparison of spectra and imaging results under different sampling conditions. (a)–(d) Spectra of SAR images. (e)–(h) Corresponding imaging results. (a) and (e) Wideband and spatially dense sampling. (b) and (f) Single-frequency and spatially dense sampling. (c) and (g) Single-frequency and spatially sparse uniform sampling. (d) and (h) Single-frequency and spatially sparse random sampling.

grating lobes in Fig. 2(h) is dispersed across the image, indicating that random sampling can effectively suppress grating lobes.

B. Principles of Diffusion Model

Diffusion models are generative models that approximate the data distribution through a parameterized Markov process consisting of a forward (diffusion) process and a reverse process. The forward process gradually adds Gaussian noise to the data via a fixed Markov chain, converting the original data distribution x_0 into a standard Gaussian distribution x_t. This process can be defined as

$$\begin{aligned} q(x_1,\ldots,x_T|x_0) &= \prod_{t=1}^{T} q(x_t|x_{t-1}) \\ &= \prod_{t=1}^{T} \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}x_{t-1},\beta_t\mathbf{I}\right) \end{aligned} \tag{7}$$

where $\mathbf{I}$ is an all-ones matrix. $\mathcal{N}$ denotes the Gaussian distribution. x_t is the latent variable at timestep $t \in 1,2,\ldots,T$. β_t represents a noise variance schedule.

A notable property of the diffusion process is that the latent variable x_t at an arbitrary timestep t can be directly sampled from x_0 and the noise variance schedule β_t, as formulated by

$$\begin{aligned} q(x_t|x_0) &= \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t}x_0,(1-\bar{\alpha}_t)\mathbf{I}\right) \\ \alpha_t &= 1-\beta_t \\ \bar{\alpha}_t &= \sum_{i=1}^{t}\alpha_i \end{aligned} \tag{8}$$

which can be further reparameterized as

$$x_t(x_0,\epsilon) = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I}) \tag{9}$$

where $\mathbf{0}$ denotes an all-zero matrix. ϵ denotes the Gaussian noise added to x_0.

The reverse process is a learnable Markov chain designed to progressively transform Gaussian noise into the target data distribution. At each timestep, it estimates the previous latent variable x_{t-1} based on current state x_t and predicted noise ϵ_θ, which can be formulated as

$$\begin{aligned} x_{t-1}(x_t,t) &= \mu_\theta(x_t,t) + \alpha_t z_\epsilon \\ &= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t,t)\right) + \alpha_t z_\epsilon \end{aligned} \tag{10}$$

where μ_θ represents the mean of the Gaussian distribution and $z_\epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I})$.

Moreover, according to [38], the objective function of the diffusion model can be formulated as follows:

$$\begin{aligned} &L_{\text{simple}}(\theta) \\ &= \mathbb{E}_{t,x_0}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right]. \end{aligned} \tag{11}$$

III. Proposed Method

A. Conditional Diffusion Model for MMW Imaging

While traditional MF-based methods can achieve high-quality imaging with wideband signals and sufficient spatial sampling rate (SR), the reconstruction problem becomes a severely ill-posed inverse problem when restricted to single-frequency and spatially sparse sampling. In such cases, the linear mapping inherent in MF-based methods inevitably leads to defocusing and severe grating lobe artifacts.

From the Bayesian perspective, the reconstruction task can be formulated as a sampling process from the posterior distribution $p(x_0|x_c)$, where x_0 is the target image, and x_c represents the conditioning information from the single-frequency and spatially undersampled measurements. However, due to the inaccessibility of the exact distributions of both x_0 and x_c, an analytical formulation of $p(x_0|x_c)$ is intractable. In this context, diffusion models provide a promising solution. As described in (10), the reverse diffusion process is determined by the predicted noise ϵ_θ, which can be conditioned on x_c by incorporating the measurement information into the noise predictor. This enables the posterior distribution

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estimation and allows the model to generate the target image x_0 through T refinement steps, which can be formulated as

$$\begin{aligned} p_\theta(x_{0:T}|x_c) &= p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t, x_c) \\ p_\theta(x_{t-1}|x_t, x_c) &= \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, x_c, t), \alpha_t^2 \mathbf{I}\right). \end{aligned} \tag{12}$$

The objective function of the diffusion model is then modified to

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t,x_0}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, x_c, t\right)\right\|^2\right]. \tag{13}$$

B. Advanced Feature Analysis on Image Phase

Direct measurement signals are infeasible as the conditional input x_c in the diffusion model due to the inherent feature discrepancy between the measurement space and the image domain. To address this, the two-stage reconstruction framework has gained increasing attention, in which a traditional MF-based algorithm, typically RMA, is first employed to generate fuzzy SAR images. This transformation effectively maps the received signals from the measurement space into the image domain, thereby providing a structurally compatible prior for subsequent DL-based refinement.

However, in the state-of-the-art methods, only the amplitude maps of the complex-value SAR images are utilized, due to their intuitive representation of object shape and scattering intensity. As analyzed in Section II-A, SAR images suffer from defocusing and severe grating lobe artifacts under single-frequency and spatially sparse sampling conditions, which degrade the quality of amplitude maps and consequently impair the final reconstruction performance. To enhance feature extraction capability under such challenging conditions, the phase information in SAR images is thoroughly explored in this section.

In real-world applications, objects can generally be regarded as single-surface structures due to the strong penetration capability of MMW signals through various nonmetallic and nonconductive materials, such as clothing, glass, and plastic. Therefore, the signal model in (1) can be reformulated to

$$\begin{aligned} s(x', y', k_0) = &\iint_D f(x_d, y_d, z_d) \\ &\times e^{-j2k_0\sqrt{(x_d-x')^2+(y_d-y')^2+z_d^2}} dx_d dy_d \end{aligned} \tag{14}$$

where D denotes the projection region on the xOy plane of the object, (x_d, y_d, z_d) represents the spatial coordinates of the object with $z_d = z_d(x_d, y_d)$ denoting its surface function, and k_0 is the wavenumber corresponding to the single-frequency signal.

Substitute (14) into (2), the reconstruction formulation expands to

$$\begin{aligned} \hat{f}(x,y,z) &= \sum_A s(x', y', k_0) e^{2k_0\sqrt{(x-x')^2+(y-y')^2+z^2}} \\ &= \iint_D \sum_A f(x_d, y_d, z_d) \\ &\quad \times e^{-j2k_0\sqrt{(x_d-x')^2+(y_d-y')^2+z_d^2}} \\ &\quad \times e^{2k_0\sqrt{(x-x')^2+(y-y')^2+z^2}} dx_d dy_d \end{aligned} \tag{15}$$

where A denotes the set of spatially sparse sampling positions, i.e., the topology of the sparse array.

By applying the Fresnel approximation, (15) can be simplified to (16), as shown at the bottom of the next page. As discussed in Section II-A, spatially random sampling can partially suppress the grating lobes resulting from the sparsity, leading to the randomly distributed topology A. Therefore, the Monte Carlo integration technique [39], [40] is employed to approximate the summation over A in (16), formulated as

$$\begin{aligned} \text{MCI} = &\sum_A e^{jk_0\frac{(x-x')^2}{z} - jk_0\frac{(x_d-x')^2}{z_d}} \\ &\times e^{jk_0\frac{(y-y')^2}{z} - jk_0\frac{(y_d-y')^2}{z_d}} \\ \approx &\frac{N_A}{L_x L_y}\iint_D e^{jk_0\frac{(x-x')^2}{z} - jk_0\frac{(x_d-x')^2}{z_d}} \\ &\times e^{jk_0\frac{(y-y')^2}{z} - jk_0\frac{(y_d-y')^2}{z_d}} dx' dy' \end{aligned} \tag{17}$$

where N_A is the number of spatial samples, and $L_x = x'_{\max} - x'_{\min}$ and $L_y = y'_{\max} - y'_{\min}$ denote the horizontal and vertical dimensions of the synthetic aperture, respectively.

Next, the principle of stationary phase (POSP) [41] is applied to evaluate the integrals with respect to x' and y', leading to the simplified form as follows:

$$\begin{aligned} \text{MCI} \approx &\frac{N_s \pi z z_d}{L_x L_y k_0 |z - z_d|} \\ &\times e^{jk_0\frac{(x-x_d)^2}{z-z_d} + jk_0\frac{(y-y_d)^2}{z-z_d} + j\frac{\pi}{2}} \\ &\times \text{Rect}\left[\frac{x-x_d}{2|z-z_d|\tan\left(\frac{\phi}{2}\right)}\right] \\ &\times \text{Rect}\left[\frac{y-y_d}{2|z-z_d|\tan\left(\frac{\phi}{2}\right)}\right] \end{aligned} \tag{18}$$

where ϕ denotes the beamwidth of the antenna elements.

Substitute (18) into (16), the imaging result becomes

$$\begin{aligned} \hat{f}(x,y,z) = &\iint_D f(x_d, y_d, z_d) \frac{N_A \pi z z_d}{L_x L_y k_0 |z - z_d|} \\ &\times e^{jk_0\frac{(x-x_d)^2}{z-z_d} + jk_0\frac{(y-y_d)^2}{z-z_d} + j2k_0(z-z_d) + j\frac{\pi}{2}} \\ &\times \text{Rect}\left[\frac{x-x_d}{2|z-z_d|\tan\left(\frac{\phi}{2}\right)}\right] \\ &\times \text{Rect}\left[\frac{y-y_d}{2|z-z_d|\tan\left(\frac{\phi}{2}\right)}\right] dx_d dy_d. \end{aligned} \tag{19}$$

Finally, by applying POSP again to solve the above integrals with respect to x_d and y_d, (19) is solved as

$$\begin{aligned} \hat{f}(x,y,z) \approx &\frac{N_A \pi^2 z z_d}{L_x L_y k_0^2} D(x,y) \\ &\times f(x, y, z_d(x,y)) e^{j2k_0(z - z_d(x,y))}. \end{aligned} \tag{20}$$

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Although (20) does not strictly represent the exact phase of SAR images due to several approximations, including the Fresnel approximation, Monte Carlo integration, and POSP, the phase features it captures remain highly reliable, as will be further demonstrated in experiments.

According to (20), within the object region D, the image phase $2k_0(z - z_d(x, y))$ is proportional to the depth difference between the object surface and the image pixel, exhibiting a continuous distribution due to the slowly varying nature of the surface. In contrast, for pixels outside the object region, the phase is primarily influenced by noise and imaging artifacts due to the weak signal response in those areas, resulting in a random distribution. Therefore, phase continuity effectively delineates object contours and can be leveraged to improve the model's boundary localization accuracy. Furthermore, the proportional relationship between the image phase and object surface depth implies that the depthwise gradient of the unwrapped phase encodes absolute surface depth, while the horizontal and vertical gradients reflect surface depth variations. Consequently, the spatial gradient of the image phase captures fine textures and 3-D structural details, enabling enhanced object recognition and detail restoration capabilities.

Moreover, considering the severely undersampled measurements, adopting 3-D imaging, as in traditional DL-based methods, would lead to significant oversampling of SAR images. In contrast, a single-depth 2-D image is information-preserving, with nearly all critical features retained, including object contours revealed by phase continuity and delicate textures and 3-D structures encoded in the phase gradient. Therefore, the 2-D SAR image at the center of the imaging region is used in subsequent discussions, substantially reducing the model complexity.

C. APIDiff Architecture

Based on the above discussion, the proposed method, APIDiff, which integrates both the amplitude and phase features to condition the diffusion model, is detailed in this section, with overall architecture illustrated in Fig. 3. First, single-slice 2-D SAR images are reconstructed through RMA at the center of the imaging region. Amplitude maps are extracted directly, while phase maps are obtained by unwrapping the original phase using the quality-guided phase unwrapping method [42]. An SW encoder is introduced to learn the prior information from amplitude and phase maps, separately. Next, a multimodal fusion module is designed to integrate these features, enhancing the representational richness. Finally, a conditional diffusion module leverages the fused features to reconstruct and refine the high-quality target image progressively. Besides, an adaptive weighting mechanism for conditional features is incorporated to accelerate model convergence.

1) SW Encoder: Artifacts arising from defocusing and grating lobes typically exhibit spatially dispersed distributions across the image, posing challenges for traditional convolutional neural network (CNN)-based architectures, which are inherently limited by their local receptive fields and struggle to capture long-range dependencies. Furthermore, CNNs tend to overemphasize low-frequency components in feature maps while underrepresenting high-frequency details [43]. This limitation is particularly detrimental in phase information extraction, where high-frequency components encode critical structural cues for object contours and surface undulations.

DWT is a widely used technique in image denoising and compression due to its ability to capture spatial and frequency information simultaneously [44]. Unlike uniform processing in the spatial domain, DWT enables separate handling of low- and high-frequency components, facilitating more precise restoration of high-frequency image details. In addition, DWT provides an effective means for extracting phase gradients by decomposing the image into horizontal, vertical, and diagonal subbands, thereby capturing directional phase variations that are closely associated with object contours, textures, and 3-D structural characteristics, leading to richer and more discriminative object representations. Moreover, since each subband has just half the resolution of the original image, convolution operations applied to these subbands effectively expand the network's receptive field with only a marginal increase in parameter count, thus enhancing global feature representation.

Therefore, DWT is incorporated into the proposed SW encoder to extract features from both spatial and frequency domains for amplitude and phase components. The SW encoder adopts a dual-branch architecture, as illustrated in Fig. 3(b). In the spatial branch, features are extracted using an RRDB [45], which employs densely connected residual paths to preserve high-frequency details and promote multilevel feature reuse, thereby improving representational capacity and training efficiency. In the wavelet branch, the input is first decomposed into four subbands via DWT, capturing image information at different frequency levels, specifically, one low-frequency component representing structural content and three high-frequency components encoding directional texture details. Each subband is processed through dedicated branches using RRDBs with shared network parameters to reduce model complexity while enabling frequency-specific feature extraction.

2) Multimodal Feature Fusion: The spatial and wavelet features extracted from amplitude and phase maps are integrated through a multimodal feature fusion module to enhance their

$$\begin{aligned}\hat{f}(x,y,z) &\approx \iint_D \sum_A f(x_d,y_d,z_d)\, e^{-j2k_0\left(z_d+\frac{(x_d-x')^2}{2z_d}+\frac{(y_d-y')^2}{2z_d}\right)} e^{j2k_0\left(z+\frac{(x-x')^2}{2z}+\frac{(y-y')^2}{2z}\right)} dx_d dy_d \\ &= \iint_D f(x_d,y_d,z_d)\, e^{j2k_0(z-z_d)} \sum_A e^{jk_0\frac{(x-x')^2}{z}-jk_0\frac{(x_d-x')^2}{z_d}} e^{jk_0\frac{(y-y')^2}{z}-jk_0\frac{(y_d-y')^2}{z_d}} dx_d dy_d \end{aligned} \tag{16}$$

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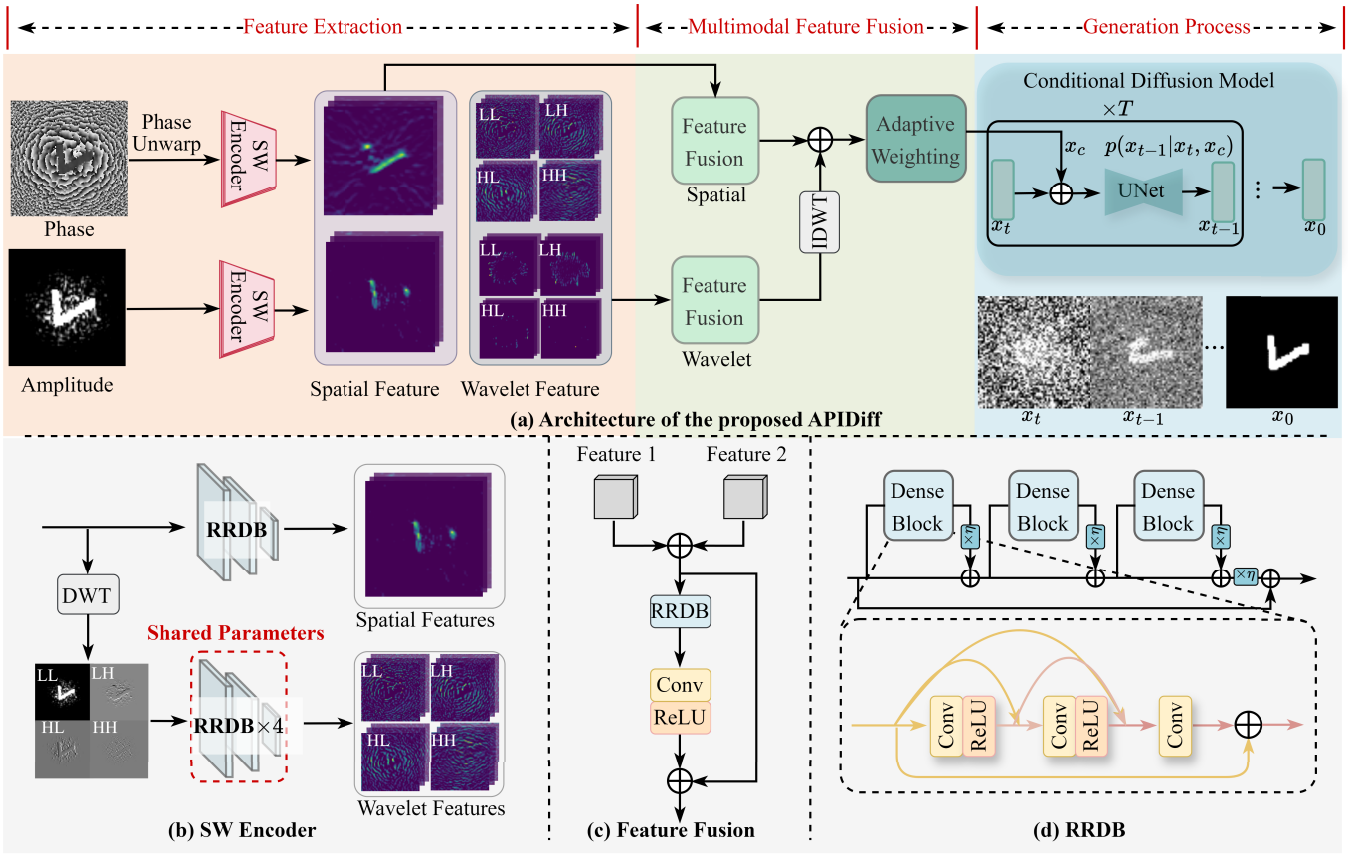

Fig. 3. Overall architecture of APIDiff. (a) Overall framework. (b) Architecture of the SW encoder. (c) Architecture of the feature fusion model. (d) Architecture of RRDB.

representational capability. Specifically, the fusion module consists of an RRDB block followed by an output layer comprising a convolutional layer and an rectified linear unit (ReLU) activation function. Spatial features from the amplitude and phase maps are fused via elementwise addition and then processed through the fusion module. Wavelet features from the four subbands are separately fused using dedicated branches and subsequently reconstructed into the spatial domain using the inverse DWT (IDWT). Finally, the spatial and wavelet features are combined through elementwise addition and fed into the diffusion model as the conditional input.

An adaptive weighting strategy is proposed, in which the conditional information is scaled by a learnable condition weight parameter γ to balance generation flexibility and fidelity. In the early stages of the reverse process, the latent variable x_t is dominated by Gaussian noise and contains limited useful information, necessitating a larger γ to provide stronger conditional guidance. As the denoising process progresses and x_t gradually converges toward the clean data x_0, the influence of the conditioning signal can be reduced by decreasing γ, thereby enhancing the model's generative flexibility. Accordingly, γ is initialized with a large value and progressively decreased during the reverse process, while being jointly optimized with the diffusion model to adaptively adjust the conditional strength across training stages. This strategy not only balances the guidance strength and diversity of the generated samples but also accelerates network convergence.

3) Conditional Diffusion Model: UNet [46] is adopted as the backbone of the conditional noise predictor due to its ability to capture hierarchical spatial features effectively. The network takes as input the randomly sampled Gaussian noise x_T, the diffusion timestep $t \in \{1, 2, \ldots, T\}$, and the condition x_c, which provides external contextual information to guide the reverse process. At each timestep t, the noise component $\epsilon_\theta(x_t, x_c, t)$ is predicted through the UNet, and subsequently utilized to compute the intermediate latent x_{t-1} from x_t according to (10). Over T iterative refinement steps, the model gradually transforms the initial Gaussian distribution into the desired conditional probability density $p(x_0|x_c)$, finally reconstructing the high-quality image x_0 through sampling.

D. Training and Inference

1) Training Process: The overall training process is detailed in Algorithm 1. The single-frequency spatially undersampled data s_s are extracted from the wideband fully sampled data s_0 using uniform-random spatial sampling patterns. The ground truth x_0 is obtained by applying RMA to s_0 over the 3-D imaging region, followed by maximum intensity projection (MIP). The first-stage reconstruction x_s is generated by applying RMA to s_s on the 2-D plane at the center of the imaging region. Amplitude and unwrapped phase maps of x_s are subsequently fed into the SW encoder and the

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Algorithm 1 Training

Input: Wideband fully sampled raw data s_0, total timestep T
Sample the single-frequency spatially undersampled data s_s
Use RMA to obtain the sparse reconstruction x_s and ground truth x_0
Perform phase unwrapping of x_s
Pre-train SW encoder and the multimodal feature fusion module
repeat
  Sample (x_s, x_0)
  Obtain the condition $x_c$from x_s
  Sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $t \sim \text{Uniform}(1, \ldots, T)$
  Perform the gradient step on $\nabla_\theta \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_t + \sqrt{1-\bar{\alpha}_t}\epsilon, x_c, t)\|$
until converged

Algorithm 2 Inference

Input: Single-frequency spatially undersampled data s_s, total timestep T
Use RMA to obtain the image x_s
Perform phase unwrapping of x_s
Obtain the condition $x_c$from x_s
Sample $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, T-1, \ldots, 1$**do**
  Sample $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$if $t > 1$, else $z = 0$
  Compute x_{t-1}according to (10):
$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, x_c, t)\right) + \alpha_t z$
end for
return x_0

multimodal feature fusion module. Due to the "divide-and-conquer" characteristic, an intermediate supervision strategy is employed to balance learning across frequency components, and the overall loss function is formulated as

$$\begin{aligned} \mathcal{L} &= \lambda_{\text{real}}\mathcal{L}_{\text{real}} + \mathcal{L}_{\text{freq}} \\ \mathcal{L}_{\text{real}} &= \|x_c - x_0\|_1 \\ \mathcal{L}_{\text{freq}} &= \sum \lambda_{\text{fi}}\mathcal{L}_{\text{fi}} \\ \mathcal{L}_{\text{fi}} &= \left\|x_{c,\text{fi}} - x_{0,\text{fi}}\right\|_1 \end{aligned} \tag{21}$$

where $\mathcal{L}_{\text{real}}$ denotes the final reconstruction loss between the ground truth x_0 and the conditional input x_c, $\mathcal{L}_{\text{freq}}$ represents the frequency-componentwise reconstruction loss over the four subbands fi in DWT with fi $\in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$, and $x_{c,\text{fi}}$ and $x_{0,\text{fi}}$ denote the reconstructed conditional input and ground truth for subband fi, respectively. The total loss is a weighted sum of the subband-specific losses $\mathcal{L}_{\text{fi}}$ and the final reconstruction loss $\mathcal{L}_{\text{real}}$, with hyperparameters λ_{real} and λ_{fi} controlling the relative contributions of different components.

After training the SW encoder and the multimodal feature fusion module under intermediate supervision, their parameters are frozen and integrated into APIDiff. Subsequently, a batch of paired samples (x_s, x_0) is drawn from the training set, where x_s is encoded to generate the conditional image x_c. The conditional noise predictor is then optimized using the gradient of the objective defined in (13), with x_t and timestep t sampled from standard Gaussian and uniform distributions, respectively.

2) Inference Process: The inference procedure is detailed in Algorithm 2. Given a single-frequency spatially undersampled measurement, the first-stage reconstruction x_s is obtained through RMA on the 2-D plane at the center of the imaging region, followed by a phase unwrapping operation to recover the continuous phase within the object region. Amplitude and unwrapped phase maps of x_s are subsequently fed into the pretrained SW encoder and the multimodal feature fusion module to extract the conditional features x_c from x_s. The sampling process of the diffusion model begins by drawing an initial noise sample $x_T \in \mathcal{N}(\mathbf{0}, \mathbf{I})$. At each timestep t, the noise prediction network estimates the added noise ϵ_θ based on the current state x_t and the condition x_c. During the reverse process, the magnitude of the predicted noise gradually decreases as t decreases. Furthermore, for $t > 1$, a random variable $z_\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is sampled and combined with the predicted noise to reconstruct x_{t-1} according to (10). When $t = 1$, z_ϵ is set to zero, resulting in x_0 as the final high-quality reconstruction.

To demonstrate the feasibility of APIDiff in practical applications, the computational complexity of its inference process is further analyzed in terms of floating-point operations (FLOPs). Both the SW encoder and the multimodal feature fusion module employ RRDBs as their basic building units. Each RRDB consists of three dense blocks (DBs), and each DB contains three convolutional layers with dense connections. Let the input and output feature maps have a spatial size of $W \times H$, an input channel dimension of C_{in}, and an output channel dimension of C_{out}. The FLOPs of each DB can be expressed as

$$\begin{aligned} \text{FLOPs}_{\text{DB}} &= KHW\left[C_{\text{in}}C_{\text{out}} + (C_{\text{in}} + C_{\text{out}})C_{\text{out}} \right. \\ &\quad \left. + (C_{\text{in}} + 2C_{\text{out}})C_{\text{out}}\right] \\ &= KHW\left[3C_{\text{in}}C_{\text{out}} + 3C_{\text{out}}^2\right] \end{aligned} \tag{22}$$

where K is a constant related to the convolution kernel settings.

Considering three DBs and one additional outer residual connection, the total computational cost of each RRDB is given by

$$\text{FLOPs}_{\text{RRDB}} = 3 \times \text{FLOPs}_{\text{DB}} + 4HWC_{\text{in}}. \tag{23}$$

Both the SW encoder and the multimodal feature fusion module consist of five RRDB units, along with either a DWT or IDWT operation. Assuming $C_{\text{in}} = C_{\text{out}} = C$ and neglecting lower order terms for simplicity, the computational costs of the SW encoder ($\text{FLOPs}_{\text{SWE}}$) and the multimodal feature fusion module ($\text{FLOPs}_{\text{MFF}}$) are approximated as

$$\begin{aligned} \text{FLOPs}_{\text{SWE}} &\approx 5 \times \text{FLOPs}_{\text{RRDB}} \approx 90KHWC^2 \\ \text{FLOPs}_{\text{MFF}} &\approx 5 \times \text{FLOPs}_{\text{RRDB}} \approx 90KHWC^2. \end{aligned} \tag{24}$$

Additionally, the UNet architecture [46] adopted in the noise predictor of the conditional diffusion model comprises four encoding and four decoding layers. Each encoding stage includes two convolutional layers followed

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by a max-pooling layer for downsampling. The computational cost of the encoder is approximated as

$$\text{FLOPs}_{\text{encoder}} \approx 16KHWC^2. \tag{25}$$

The decoder mirrors the encoder structure and employs bilinear interpolation for upsampling and concatenation with features from the encoder, resulting in a total computational cost of

$$\text{FLOPs}_{\text{decoder}} \approx 24KHWC^2. \tag{26}$$

Consequently, the overall computational complexity of the UNet can be approximated as

$$\begin{aligned}\text{FLOPs}_{\text{UNet}} &\approx \text{FLOPs}_{\text{encoder}} + \text{FLOPs}_{\text{decoder}} \\ &\approx 40KHWC^2.\end{aligned} \tag{27}$$

Since the reverse process involves T iterative timesteps, the total computational load of APIDiff can be formulated as

$$\begin{aligned}\text{FLOPs}_{\text{total}} &\approx 2 \times \text{FLOPs}_{\text{SWE}} + \text{FLOPs}_{\text{MFF}} \\ &\quad + T \times \text{FLOPs}_{\text{UNet}} \\ &\approx 40TKHWC^2.\end{aligned} \tag{28}$$

For a typical configuration ($K = 9$, $T = 20$, $H = W = 80$, and $C = 32$), the total computational cost is approximately 47.2 GFLOPs, which is computationally feasible on modern GPUs and ensures the applicability of APIDiff in real-time imaging.

IV. Experiment and Analysis

In this section, extensive simulations and real-world experiments are conducted to validate the effectiveness of the proposed method in comparison with state-of-the-art techniques. Ablation studies are further performed to evaluate the contribution of each component within the proposed framework. Additionally, experiments under varying SRs and signal-to-noise ratios (SNRs) are carried out to assess the method's robustness.

A. Simulation and Real-World Dataset

1) Simulation Dataset: As stated in Section III-D, although 2-D imaging is sufficient for generating the first-stage reconstruction x_s, 3-D imaging using wideband and spatially densely sampled measurements is required to generate the ground truth x_0. A simulated MMW 3-D imaging system, employing wideband signals and spatially dense sampling, is constructed for numerical verification, with geometry consistent with Fig. 1. The signal frequency ranges from 24.4 to 27.8 GHz, comprising 60 equidistant samples. The synthetic aperture measures 0.4 ×0.4 m, sampled at a 5-mm interval. The 3-D imaging region spans 0.4 × 0.4 ×0.3 m, with $80 \times 80 \times 60$ equidistant sampling points. The MNIST dataset [47] of handwritten digits is used as the test specimens, with each digit positioned at the center of the 3-D imaging region.

As the network input, the single-frequency spatially undersampled data are extracted from the wideband densely sampled data at 27.3 GHz using uniform-random spatial sampling patterns. In comparative and ablation studies, the spatial SR of the single-frequency data is set to 50%. In extended experiments, the SR varies from 25% to 75% in increments of 25%. A total of 5000 simulated samples are generated, with 4500 allocated for training and 500 for testing and validation. Representative visualization samples derived from the MNIST dataset are shown in Fig. 4, illustrating the defocusing and grating lobe artifacts discussed in Section II-A. As the SR increases, the quality of the amplitude map deteriorates more severely.

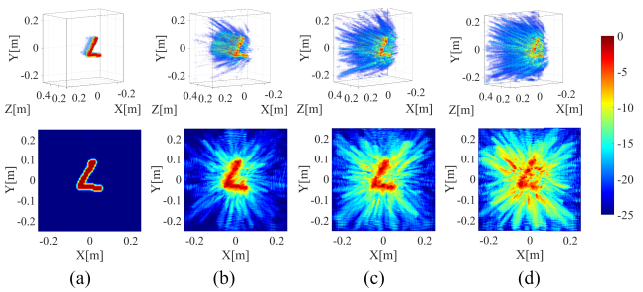

Fig. 4. Representative visualization samples generated from the MNIST dataset, where the first and second rows depict the 3-D rendering results and the MIP results of the reconstructed images of the digit "7," respectively. (a) Ground truth from wideband fully sampled imaging. (b) Results obtained from single-frequency and 75% spatially undersampled data. (c) Results obtained from single-frequency and 50% spatially undersampled data. (d) Results obtained from single-frequency and 25% spatially undersampled data.

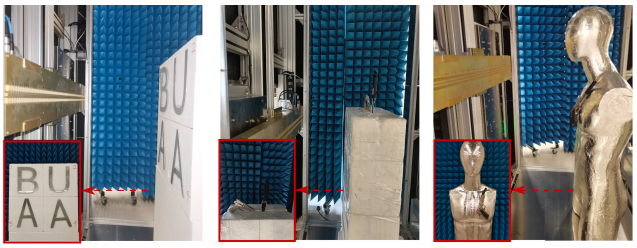

Fig. 5. Experimental system setup. (a) 3-D-printed "BUAA." (b) Gun and knife. (c) Human body model.

2) Real-World Dataset: Similar to the simulation setup, a larger scale imaging system is employed to acquire the measured dataset, as shown in Fig. 5. The synthetic aperture measures 0.96 ×1.8 m, and the 3-D imaging region spans 0.96 × 1.8 ×0.3 m. All other parameters remain consistent with those of the simulated dataset.

The dataset contains six categories of objects with varying materials and sizes, including 3-D-printed letters, guns, knives, scissors, axes, and human body models. A total of 5528 images are collected, with 4500 used for training and the remaining 1028 reserved for testing and validation. The distribution of object categories in the dataset is presented in Fig. 6.

It is worth noting that due to the incomplete consistency between the simulation and real-world datasets in terms of parameter settings, the training and inference processes are conducted independently on each dataset without mixed training.

B. Experimental Setting

For the SW encoder and the multimodal feature fusion module, the wavelet basis of the DWT is set to the "Haar" wavelet

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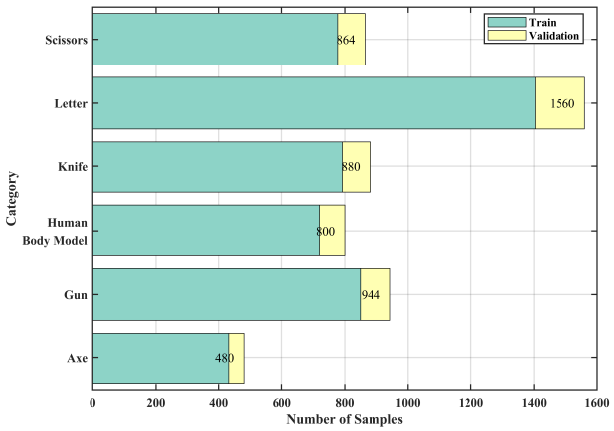

Fig. 6. Statistics of various objects in the real-world dataset.

due to its computational efficiency and simplicity [44]. The loss weight coefficients $\{\lambda_{\text{real}}, \lambda_{\text{fi}}\}$ are empirically determined through grid search and set to $(5, 1, 2, 2, 4)$ to achieve optimal performance. The effectiveness of the intermediate supervision loss and the choice of λ_{real} and λ_{fi} are further discussed in the ablation study.

For the conditional diffusion model, a cosine noise schedule [48] with a small offset of 0.008 is adopted to prevent the noise level from approaching zero as $t \to 0$. The total number of diffusion timesteps is set to $T = 20$, and the influence of different T values on reconstruction performance is analyzed in the ablation study. Considering the tradeoff between model complexity and reconstruction accuracy, the hidden dimension of feature maps is set to 32, while the convolution kernel size is set to 3 with a stride of 1 and padding of 1.

During training, the batch size is set to 16, and the Adam optimizer is employed with an initial learning rate of 1×10^{-4}, which is gradually decayed to enhance convergence stability. On both the simulated and measured datasets, the SW encoder and the multimodal feature fusion are trained for 50 epochs, while the conditional diffusion model is trained for 200 epochs to ensure full convergence.

All implementations are based on PyTorch [49] and executed on an Ubuntu 20.04.6 LTS system. The experiments are conducted on a workstation equipped with a 4.2-GHz AMD Ryzen 9 7950X3D CPU and an NVIDIA RTX 4090 GPU.

C. Evaluation Matrix and Comparative Methods

To quantitatively evaluate the reconstruction performance of the proposed model, three widely used image quality assessment metrics, including peak SNR (PSNR), structural similarity index measure (SSIM) [50], and learned perceptual image patch similarity (LPIPS) [51], are employed to compare the fidelity of reconstructed images obtained by different methods.

PSNR is a logarithmic representation of the inverse of the mean squared error (mse), scaled by the maximum possible pixel intensity MAX_I of the input image. Mathematically, PSNR is defined as

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I{}^2}{\text{mse}} \right). \tag{29}$$

SSIM is a multidimensional image quality metric that evaluates the perceptual similarity and structural distortion between two images. It captures differences in luminance, contrast, and structural information, thereby reflecting changes in underlying image structures. SSIM is defined as

$$\text{SSIM}(A, B) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x{}^2 + \mu_y{}^2 + c_1)(\sigma_x{}^2 + \sigma_y{}^2 + c_2)} \tag{30}$$

where μ_A and μ_B denote the mean intensities of images A and B, respectively, σ_A^2 and σ_B^2 represent their variances, σ_{AB} is the covariance between A and B, and c_1 and c_2 are small constants introduced to stabilize the denominator.

LPIPS is a DL-based image quality metric that models the perceptual mechanisms of the human visual system, making it particularly suitable for evaluating complex and textured images. Compared to traditional metrics, LPIPS demonstrates higher sensitivity to subtle distortions in structural details, delicate textures, and semantic features.

To validate the effectiveness of the proposed method, comparisons are conducted against several baseline approaches, including an MF-based algorithm, RMA, a CS-based method, FISTA [22], and a DL-based end-to-end reconstruction network, Deep2S [33]. Additionally, our previous work, SFADiff [14], is also included for comparison. For a fair comparison, the parameters of FISTA are carefully tuned to achieve optimal PSNR performance. Specifically, the number of iterations is set to 20, while the hyperparameters are adjusted to their optimal values. The Deep2S and SFADiff models are trained using the same datasets and experimental settings as the proposed method. Further details on the implementation of Deep2S and SFADiff can be found in their original publications.

D. Performance

1) Experiments on Simulated Dataset: The ground truth and imaging results of different methods at SRs of 75%, 50%, and 25% are presented in Fig. 7, where the ground truth is reconstructed using the RMA with wideband and spatially densely sampled measurements. It can be observed that the imaging results of RMA in the second column suffer from severe degradation. The absence of range resolution due to single-frequency measurements leads to blurring and defocusing, completely filling the hollow region in the digit "8." The violation of the Nyquist sampling criterion introduces pronounced grating-lobe artifacts, significantly reducing the dynamic range of the images. Although FISTA can partially suppress these artifacts, the blurring and defocusing issues remain unresolved, due to the extreme sparsity of the single-frequency spatially undersampled measurements and the limited capability of CS-based methods to extract and exploit structural features. Moreover, its imaging quality deteriorates significantly as the SR decreases, further rendering reconstruction infeasible under low-SR conditions. In contrast, Deep2S and SFADiff produce suboptimal imaging

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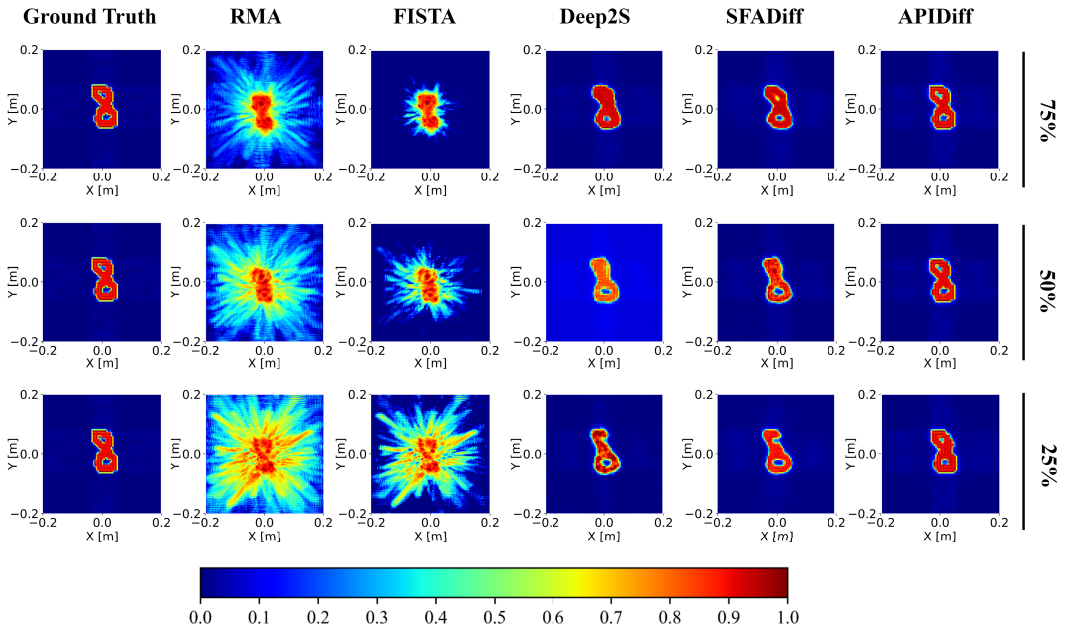

Fig. 7. Comparison of imaging results obtained by different methods on the simulated dataset at spatial SRs of 75%, 50%, and 25%.

TABLE I
COMPARATIVE QUANTITATIVE ANALYSIS OF METHODS ACROSS DIFFERENT SRS BASED ON PSNR, SSIM, AND LPIPS

Methods	Metrics	Spatial SR			
		75%	50%	25%	Avg.
RMA	PSNR(dB)	17.5882	14.3481	11.6628	14.5330
	SSIM	0.1752	0.1191	0.0779	0.1241
	LPIPS	0.3538	0.4810	0.5507	0.4618
FISTA	PSNR(dB)	23.2985	20.4541	17.8714	20.5413
	SSIM	0.6944	0.6360	0.3765	0.5690
	LPIPS	0.1123	0.1241	0.4342	0.2235
Deep2S	PSNR(dB)	33.6257	29.7839	24.9214	29.4437
	SSIM	0.9717	0.9636	0.9384	0.9579
	LPIPS	0.0077	0.0161	0.0428	0.0222
SFADiff	PSNR(dB)	35.8525	30.4878	25.6171	30.6525
	SSIM	0.9832	0.9744	0.9492	0.9689
	LPIPS	0.0042	0.0122	0.0332	0.0165
APIDiff	PSNR(dB)	**37.9128**	**34.3422**	**27.8842**	**33.3797**
	SSIM	**0.9909**	**0.9881**	**0.9729**	**0.9840**
	LPIPS	**0.0026**	**0.0060**	**0.0194**	**0.0093**

results across varying SRs. However, they fail to recover fine structures and edge details, particularly under low SR conditions, due to degraded amplitude maps. Compared to all competing methods, APIDiff achieves the highest image quality and successfully restores detailed information even at extremely low SRs, demonstrating the effectiveness of the proposed method.

The quantitative comparison of different methods is presented in Table I, with the best results highlighted in bold. It can be observed that DL-based methods outperform the traditional MF-based (RMA) and CS-based (FISTA) methods across all spatial SRs, demonstrating the effectiveness of DL-based approaches for MMW sparse reconstruction. Among these methods, the proposed APIDiff achieves the highest imaging performance across all evaluation metrics. Compared to the best results achieved by competing methods, APIDiff yields an average improvement of 2.73 dB in PSNR, 1.56% in SSIM, and a 43.49% reduction in LPIPS across the three spatial SRs. Furthermore, although all evaluation metrics degrade as the SR decreases due to the reduced amount of available measurement information, APIDiff exhibits relatively minor performance degradation, with an average drop of only 10.0286 dB in PSNR, 0.0180 in SSIM, and an increase of 0.0168 in LPIPS. These results confirm the high stability and robustness of the proposed method, enabling effective image reconstruction even under extremely low SR conditions.

2) Experiments on Real-World Datasets: Fig. 8 shows the ground truth and imaging results for "BUAA" and concealed weapons obtained by different methods under various SRs, with observed phenomena consistent with those in the simulation dataset. The RMA results are severely defocused and suffer from strong artifacts. FISTA can partially suppress artifacts and achieve barely acceptable reconstruction at 75% SR. Its performance deteriorates rapidly as the SR decreases, due to its reliance on manually designed sparsity priors and limited representational capacity for image features. Compared to traditional methods, DL-based approaches exhibit superior imaging performance, due to their powerful feature learning capabilities. However, for Deep2S and SFADiff, issues such as target loss and image blurring are also evident. Specifically, the characters "U" and "A" in "BUAA" cannot be fully reconstructed, and structural details in the imaging results of the knife and gun are lost. These issues become progressively more severe as the SR decreases, attributed to the models' heavy dependence on the quality of the first-stage reconstruction and the severely degraded amplitude maps.

In contrast, the proposed APIDiff achieves the highest imaging quality, effectively preserving fine structures, delicate textures, and precise edge contours even under extremely

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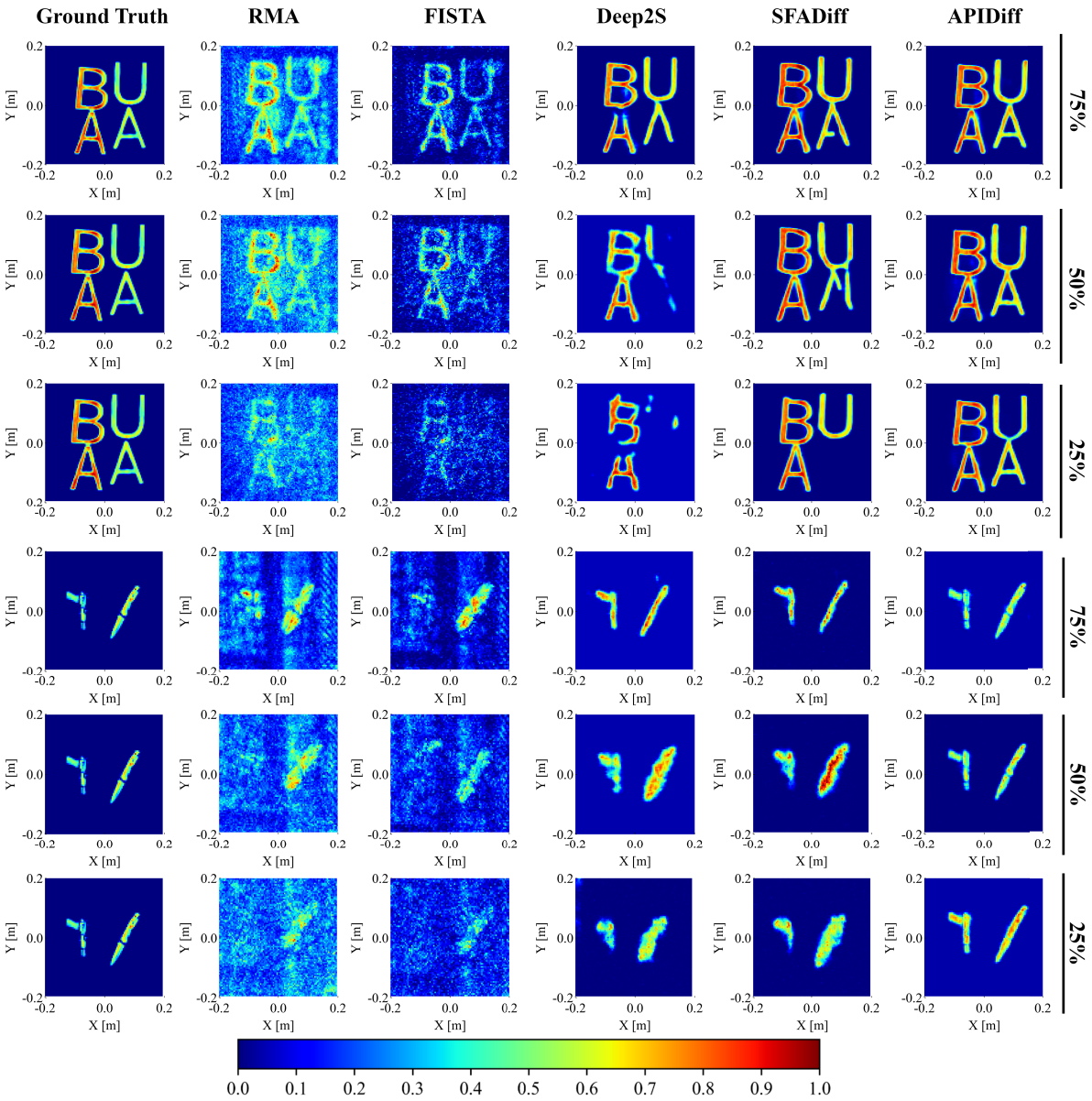

Fig. 8. Imaging results for "BUAA" and concealed objects obtained with different methods under various spatial SRs. Rows 1–3 correspond to "BUAA," and rows 4–6 correspond to concealed objects. Besides, rows 1 and 4 present results at 75% spatial SR, rows 2 and 5 at 50% spatial SR, and rows 3 and 6 at 25% spatial SR.

low SR conditions. This superior performance is primarily attributed to the incorporation of phase information, which provides additional priors on edge contours and 3-D structural features. Moreover, the unwrapped phase maps of the imaging results under single-frequency and spatially sparse sampling conditions are illustrated in Fig. 9. It can be observed that the unwrapped phase remains continuous within the object region and varies consistently with target surface undulations, thereby validating the analysis presented in Section III-B.

The imaging results for the human body model are further presented to validate the effectiveness of APIDiff, as illustrated in Fig. 10. Compared with the previously evaluated objects, the human body model exhibits larger surface undulations, greater depth range, and more complex 3-D structures, making the reconstruction task more challenging. As shown in Fig. 10(a), RMA, FISTA, and Deep2S can only recover coarse shapes of partial regions of the human body model. SFADiff generates a low-resolution or fuzzy image of the entire human body model by leveraging the diffusion model to learn richer image priors. However, its reliance solely on degraded amplitude information still leads to loss of structural details. In contrast, the proposed APIDiff achieves high-quality reconstruction, with results closely resembling the ground truth, successfully recovering the complete target while preserving fine image details. The unwrapped phase map of the imaging result for the human body model is also shown in Fig. 10(b), where the continuous region precisely matches the target boundary and the gradient aligns with the depth variation of the target surface, consistent with the earlier analysis. For a more intuitive comparison, the horizontal and vertical intensity profiles of

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TABLE II
QUANTITATIVE EVALUATION OF IMAGING PERFORMANCE ON REAL-WORLD DATASET ACROSS DIFFERENT SPATIAL SRS

Methods	Metrics	General Object				Huamn Body Model			
		Spatial SR				Spatial SR			
		75%	50%	25%	Avg.	75%	50%	25%	Avg.
RMA	PSNR(dB)	12.0613	10.9009	10.7104	11.2242	21.3833	19.8044	16.3835	19.8571
	SSIM	0.0561	0.0356	0.0224	0.0380	0.3228	0.0930	0.0219	0.1459
	LPIPS	0.8174	0.9081	0.9321	0.8859	0.5126	0.6685	0.8186	0.6666
FISTA	PSNR(dB)	19.9582	18.2517	18.0753	18.7617	21.5631	21.1336	18.4138	20.3702
	SSIM	0.5803	0.4649	0.3683	0.4712	0.7224	0.3945	0.0580	0.3916
	LPIPS	0.5479	0.6677	0.7562	0.6573	0.3800	0.5521	0.7784	0.5702
Deep2S	PSNR(dB)	27.6126	26.5316	21.3859	25.8434	22.0696	21.5921	20.7503	21.4707
	SSIM	0.9313	0.9127	0.9077	0.9172	0.8345	0.8164	0.6718	0.7742
	LPIPS	0.0819	0.0965	0.1656	0.1147	0.1986	0.2858	0.3811	0.2885
SFADiff	PSNR(dB)	29.4746	28.3041	24.8191	27.5326	23.1082	22.7498	22.0636	22.6405
	SSIM	0.9361	0.9255	0.9162	0.9259	0.8644	0.8477	0.7882	0.8334
	LPIPS	0.0535	0.0604	0.0549	0.0563	0.1299	0.1329	0.1721	0.1449
APIDiff	PSNR(dB)	**31.1670**	**29.8462**	**25.5921**	**28.8684**	**23.9701**	**23.4453**	**22.3849**	**23.2668**
	SSIM	**0.9522**	**0.9442**	**0.9317**	**0.9427**	**0.9070**	**0.8958**	**0.8907**	**0.8978**
	LPIPS	**0.0434**	**0.0503**	**0.0513**	**0.0483**	**0.1060**	**0.1121**	**0.1247**	**0.1143**

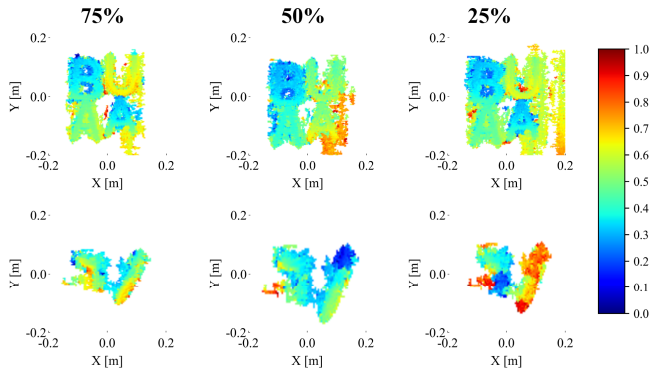

Fig. 9. Unwrapped phase maps of "BUAA" and concealed weapons under various spatial SRs.

the imaging results obtained by different methods are depicted in Fig. 10(c), respectively. The profiles produced by APIDiff exhibit closer agreement with the ground truth, demonstrating that APIDiff can recover finer structural details and more accurate edge information than the compared methods.

The quantitative evaluation results are presented in Table II, with the comparison consistent with those observed in the simulation dataset. APIDiff outperforms all other comparative methods across the three evaluation metrics under different SRs, and its metric variations across SR levels are relatively slight, demonstrating the strong stability and robustness of APIDiff. Specifically, for general objects, APIDiff achieves an average improvement of 1.34 dB in PSNR, 1.8% in SSIM, and a 14.05% reduction in LPIPS. Moreover, APIDiff achieves superior reconstruction performance for the human body model, with an average gain of 0.63 dB in PSNR, 7.87% in SSIM, and a 20% reduction in LPIPS compared to other methods. This performance advantage arises from the integration of phase features, combined with the robust feature extraction capability of the SW encoder and multi-modal feature fusion module.

TABLE III
INFERENCE TIME OF DIFFERENT METHODS

Method	RMA	FISTA	Deep2S	SFADiff	APIDiff
Time (s)	0.01	0.44	0.10	0.31	0.12

To further demonstrate the feasibility of APIDiff in practical applications discussed in Section III-D, the inference times of different methods under identical hardware and input conditions are presented in Table III. Among all approaches, RMA achieves the shortest inference time of only 0.01 s, attributed to its concise algorithmic structure and efficient acceleration via the fast Fourier transform. FISTA exhibits the longest inference time of 0.44 s, resulting from its reliance on multiple iterative steps. In contrast, DL-based methods achieve faster inference due to the benefits of offline training. Taking both imaging quality and computational efficiency into account, APIDiff achieves superior reconstruction performance while requiring only 0.12 s for reconstruction, making it a highly suitable solution for practical applications.

E. Ablation Study

In this section, ablation studies are conducted to evaluate the contributions of phase features and wavelet features to reconstruction quality under single-frequency and spatially sparse sampling conditions. Additionally, the effects of different loss function weight settings and the impact of varying numbers of diffusion timesteps on reconstruction performance are investigated.

1) Effects of Phase Features and Wavelet Features: The influence of phase and wavelet features on APIDiff's performance is assessed by training three additional models on the real-world dataset with a fixed spatial SR of 50%. In the baseline model, both phase and wavelet features are disabled. The

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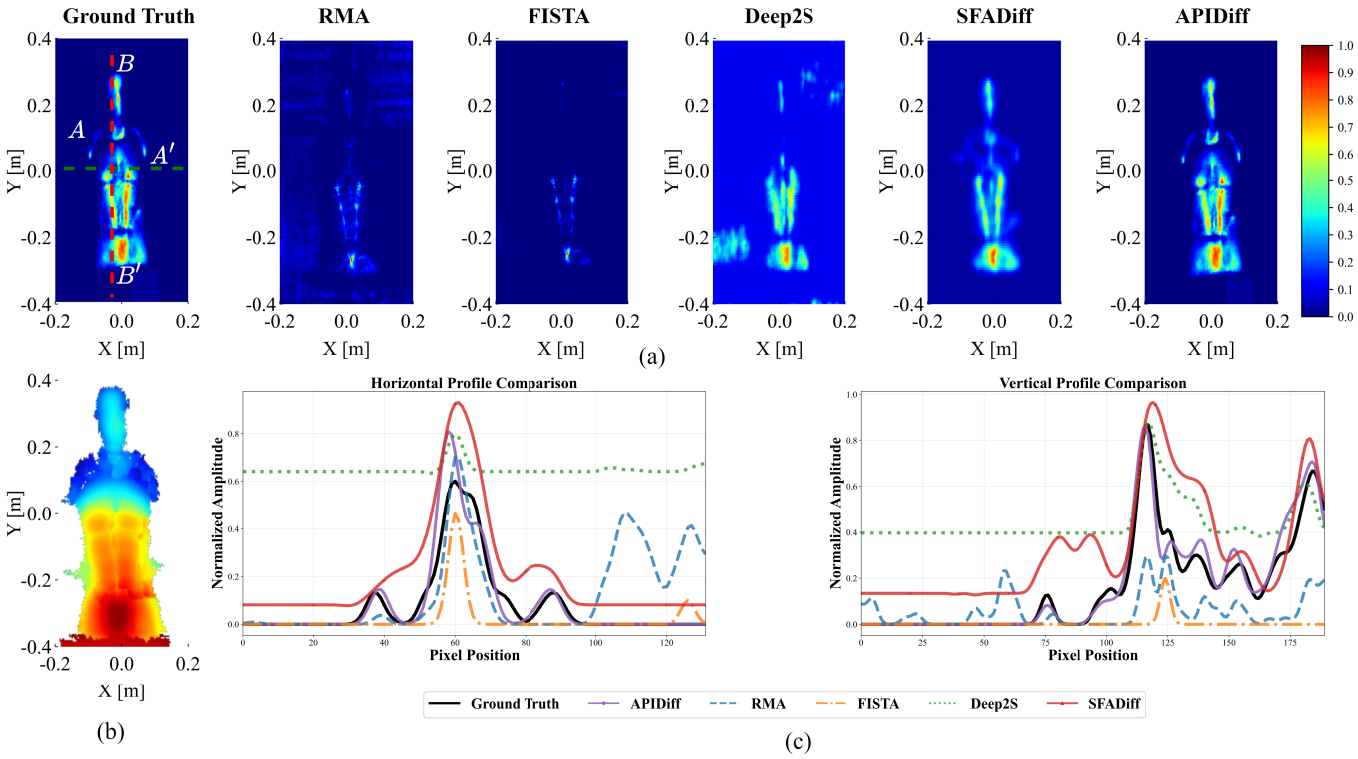

Fig. 10. Comparison of different reconstruction methods for the human body model at 75% spatial SR. (a) Reconstructed images obtained by different methods. (b) Unwrapped phase map of the imaging result for the human body model. (c) Horizontal and vertical intensity profiles extracted along lines A–A' and B–B'.

TABLE IV
QUANTITATIVE RESULTS OF THE ABLATION STUDY

Phase	DWT	PSNR(dB)	SSIM	LPIPS
×	×	26.9169	0.8953	0.0882
✓	×	28.0442	0.9269	0.0720
×	✓	27.7725	0.9267	0.0725
✓	✓	**29.8462**	**0.9442**	**0.0503**

TABLE V
PERFORMANCE COMPARISON OF SW ENCODER AND MULTIMODAL FEATURE FUSION MODULE UNDER DIFFERENT LOSS FUNCTION WEIGHTS

Weights	PSNR (dB)	SSIM	LPIPS
(1, 0, 0, 0, 0)	20.1730	0.8799	0.2240
(1, 1, 1, 1, 1)	22.1211	0.8914	0.2006
(5, 1, 1, 1, 1)	22.1780	0.9088	0.1981
(5, 1, 2, 2, 2)	23.0773	0.9101	0.1699
(5, 1, 2, 2, 4)	**23.1092**	**0.9179**	**0.1549**

second and third models extend the baseline by incorporating phase features or wavelet features individually, enabling an evaluation of their respective contributions to reconstruction performance. Quantitative results are presented in Table IV. Compared to the baseline, integrating phase information yields improvements of 1.1273 dB in PSNR, 0.0316 in SSIM, and a 0.0162 reduction in LPIPS. This enhancement is attributed to the additional priors provided by phase features on edge contours and 3-D structures. Similarly, introducing wavelet features leads to gains of 0.8556 dB in PSNR, 0.0314 in SSIM, and a 0.0157 reduction in LPIPS, due to their ability to capture information across multiple frequency subbands, which enables effective artifact suppression while preserving high-frequency details. Furthermore, combining both phase and wavelet features results in a significant improvement of 2.9293 dB in PSNR, 0.0489 in SSIM, and a 0.0379 reduction in LPIPS, demonstrating their complementary roles in enhancing APIDiff's imaging performance.

2) Effects of Loss Function Weights in Intermediate Supervision Strategy: The loss function weight coefficients in the intermediate supervision strategy in (21) are empirically set to $(5, 1, 2, 2, 4)$, following the principles outlined in the following. First, all loss terms are maintained at comparable scales to ensure balanced optimization. Second, the final reconstruction loss $\mathcal{L}_{\text{real}}$ is assigned the highest weight to preserve data fidelity and overall structural accuracy. Third, the high-frequency components in $\mathcal{L}_{\text{fi}}$ are given moderately higher weights than the low-frequency component to prevent the model from overemphasizing easily reconstructed flat regions while neglecting complex textures and details.

Table V presents the quantitative comparison between the conditional input x_c and the ground truth x_0 under different loss weight configurations for training the SW encoder and multimodal feature fusion module. The five coefficients correspond to the five weights defined in (21): λ_{real} and four λ_{fi} with $\text{fi} \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$. A comparison of the results under weight configurations (1, 0, 0, 0, 0) and (1, 1, 1, 1, 1) indicates that incorporating intermediate supervision

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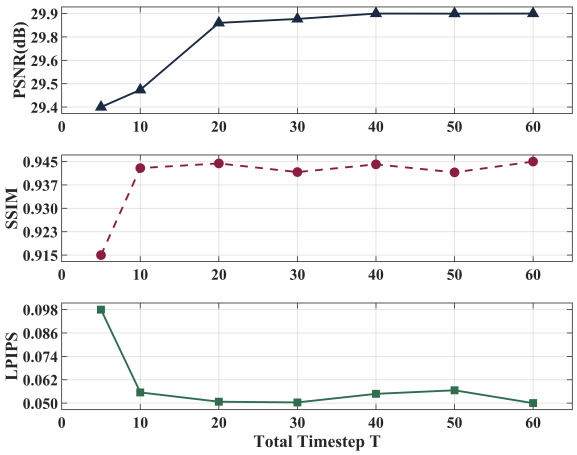

Fig. 11. Quantitative evaluation of the imaging results with different numbers of timesteps.

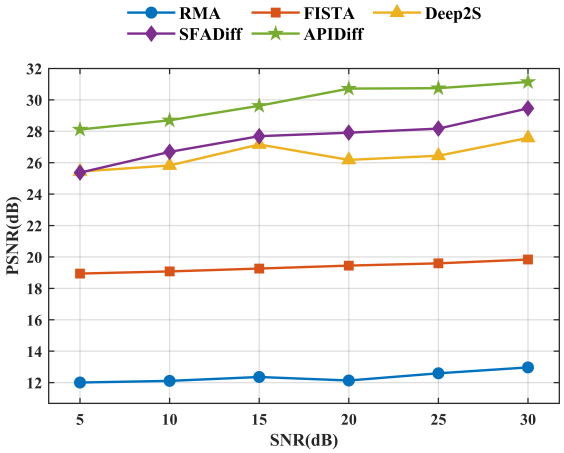

Fig. 12. Performance evaluation of different methods under varying SNRs based on PSNR.

significantly enhances reconstruction performance, yielding a 2.0-dB improvement in PSNR, a 1.3% increase in SSIM, and a 10% reduction in LPIPS. Comparing configurations (1, 1, 1, 1, 1) and (5, 1, 1, 1, 1) reveals that increasing the weight of the final reconstruction loss $\mathcal{L}_{\text{real}}$ further improves reconstruction accuracy and structural consistency. Furthermore, the results from configurations (5, 1, 2, 2, 2) and (5, 1, 2, 2, 4) demonstrate that assigning moderately higher weights to the high-frequency subbands leads to improved reconstruction accuracy, as fine textures and edge details are more effectively preserved, resulting in the best overall performance.

3) Effects of Different Numbers of Timesteps: Fig. 11 illustrates the evolution of quantitative metrics for APIDiff across varying numbers of timesteps. The experiments are performed using the same dataset and setup as in the first ablation study. As the number of timesteps T increases, PSNR and SSIM generally improve, while LPIPS decreases, indicating progressively better reconstruction quality. Notably, when $T > 20$, performance gains become marginal, suggesting diminishing returns from additional timesteps. Considering both reconstruction accuracy and computational efficiency, T is set to 20 to achieve an optimal tradeoff between image quality and inference time.

F. Extensions

To validate the robustness of APIDiff against degraded inputs, the imaging performance under different SNRs is evaluated. The extended experiments are conducted on the real-world dataset with the spatial SR fixed at 50%, ensuring the applicability of the results to practical scenarios. Fig. 12 illustrates the variation of PSNR for different methods under varying SNR conditions. It can be observed that traditional imaging methods, RMA and FISTA, are least affected by SNR variations because the severe artifacts introduced by single-frequency and spatially sparse sampling dominate the overall image degradation, thereby reducing the relative impact of noise. In contrast, APIDiff exhibits the smallest PSNR degradation among DL-based methods as SNR decreases, indicating its strong robustness under challenging conditions.

V. Conclusion

In this article, an APIDiff model is proposed to address the challenge of reconstructing high-quality MMW images with single-frequency and sparse array configurations. Amplitude and phase features of SAR images obtained using traditional MF-based methods are incorporated into a conditional diffusion framework, where amplitude maps provide information on the target's coarse 2-D shape and scattering intensity, while phase maps encode the target's contours, textures, and 3-D structural characteristics, thereby facilitating more accurate recovery of target details compared to traditional DL-based methods. An SW encoder is developed based on the DWT to extract multimodal representations from both spatial and frequency domains, enabling comprehensive exploitation of complementary amplitude and phase features. A multimodal fusion module is then introduced to effectively integrate these features, providing more informative conditional guidance for the image generation. Furthermore, an adaptive weighting strategy is employed to accelerate model convergence by dynamically adjusting the contribution of conditional information across generation timesteps. The experimental results on both simulated and real-world datasets demonstrate that the proposed APIDiff achieves superior imaging quality compared to state-of-the-art methods across varying target types, spatial SRs, and SNR levels, highlighting its excellent performance and robustness.

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